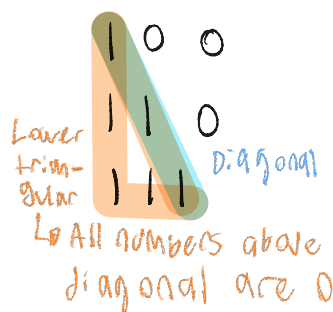
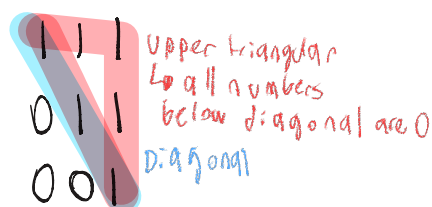
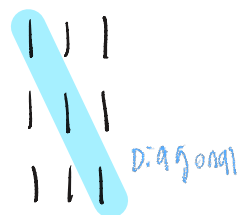


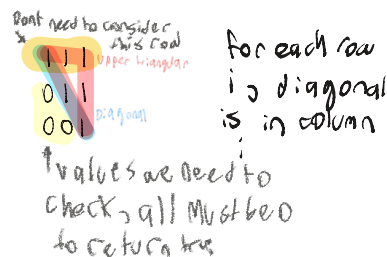
Transpose
 $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Number of cols becomes Number of rows
 and number of rows becomes number of columns

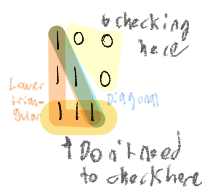
Matrices



is Upper Triangular



is Lower Triangular



is Magic

What we need to check:

↳ Matrix is square

↳ $n \text{ rows} = n \text{ cols}$

↳ all rows sum to the same #

↳ Find initial sum and check if other rows have the same sum

↳ all cols sum to that same #

↳ Check if all col sums match initial sum

↳ both diagonals sum to that same #

↳ Sums of main and left diags both = initial sum

↳ x is $n \times n$, contains all values $[1, n^2]$ exactly once

$\begin{bmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{bmatrix}$ is $\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
 True if we have seen a # before in a prev. row

Tracking exactly Once

index = 0 1 2 3
 1 2 [F F F F]
 3 4 Value 2 2 3 4
 First row
 index = 0 1 2 3
 [T T F F]
 Value 2 2 3 4

Booleans

initialized to false in Java by default

Second row
 Have we seen 3? → No, it's F
 Have we seen 4? → No, it's F
 index = 0 1 2 3
 [T T T T]
 Value 2 2 3 4